Children First, Housing Later*

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October 20, 2026

Abstract

Should housing assistance require sobriety? A general-equilibrium model of homelessness, drug use, drug prices and mortality is calibrated to US data. Use concentrates among low-income households because a substance use disorder's mortality risk, unlike its unemployment risk, is uninsurable. Drug use harms users' children, cutting their consumption and, through a parent's disorder, their adult earnings. In the long run, sobriety conditions reduce these harms and, in aggregate, also childhood homelessness. Consequently, a newborn's welfare rises by 2.57 percent of lifetime consumption, of which three quarters is an externality its parent does not count, and in general equilibrium parents gain too. Even along the transition, homelessness and childhood homelessness do not rise, and newborns still gain. The model reproduces micro evidence of randomised controlled trials, and users with a disorder lose, but macro effects on other households and children overturn the verdict. Moral hazard alone does not justify conditionality.

Keywords: homelessness; housing vouchers; substance use disorder; intergenerational spillovers; externalities.

JEL classification: I18; I38; H23; R21; J62.

1 Introduction

According to the point-in-time (PiT) count of January 2024 performed by the Department of Housing and Urban Development (HUD), at least 771,480 people were sleeping in shelters or on the street, the most since the count began. In July 2025 an executive order directed the federal agencies to end support for Housing First policies that, in its words, "fail to promote treatment, recovery, and self-sufficiency".[1] Housing First gives low-income households housing assistance with no requirement to stop using drugs; the alternative, Treatment First, makes the assistance conditional on sobriety and, for those with a substance use disorder (SUD), on being in treatment as well. The funding notice that HUD issued after the order would have capped permanent housing, through which most Housing First assistance is given, at 30 percent of renewal funding, down from more than 80 percent, and would have opened the rest to transitional housing that can require treatment and sobriety, if not for a federal court striking the notice down.[2]

*I thank Ayşe İmrohoroğlu, Kai Zhao, Joseph Steinberg, Efraim Benmelech, Jonathan Alschech, Ryan Quandt, Christopher Sharp and the seminar participants at the University of Toronto for their helpful comments. The data and replication code are available on GitHub at `https://github.com/srotberg/children_first_housing_later`.

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[1] Executive Order 14321, *Ending Crime and Disorder on America's Streets*, 24 July 2025, *Federal Register* 90, 35817–35820.

[2] HUD, *FY 2025 Continuum of Care Competition and Youth Homeless Demonstration Program Grants*, FR-6900-N-25, 13 November 2025; on the shares of funding, Congressional Research Service, Insight IN12626. The ruling is *State of Washington v. U.S. Department of Housing and Urban Development*, No. 1:25-cv-00626 (District of Rhode Island, 29 June 2026), which set aside both the November and the December 2025 notices. See `https://www.npr.org/2025/11/14/nx-s1-5553561/homelessness-housing-funding-trump-administration-hud`; on the court's ruling, `https://marylandmatters.org/2026/06/30/ri-federal-judge-strikes-down-huds-2025-restrictions-for-housing-first-initiatives/`.

The debate between Housing First and Treatment First is usually framed as a contest between two behavioural claims about the households who receive the assistance.[3] Critics of Housing First say that unconditional assistance creates moral hazard: it weakens the incentive to stay sober and employed, so drug use and SUD rise, and with them homelessness. The result is a vicious circle, whose cost falls on others who fund the programme. Critics of Treatment First say that conditionality delays stable housing for those who need it most, because recovering while homeless is harder. In their argument the circle runs the other way, from homelessness to more SUD and back, so Housing First in fact improves the budget. Randomised trials of Housing First speak to both claims: they find large gains in housing stability and mixed-to-null effects on drug use (Section 2). However, they follow the participants for at most three years, and none follows the children into adult life, tracks households outside the trial, or measures the effect on the government's budget, and only one small trial, of homeless adults already in treatment, compares a sobriety condition with unconditional housing (Milby et al., 2005). A model can do all four.

I weigh in on the debate by building a quantitative life-cycle model in which households decide whether to use drugs knowing the risk of acquiring a SUD, and in which drug use harms not only the user but also the user's child. I begin by showing that the moral hazard that dominates the debate does not drive the answer at all: I remove the externalities that harm children, leaving only the change in behaviour and its impact on the budget. A newborn is then worse off in the long run under Treatment First, by 1.09 percent in consumption-equivalent terms, even when drug use and SUD fall substantially and what the drug test saves on vouchers, together with the higher revenues from more employment, is spent on more vouchers for other households. This is because, without externalities, conditionality removes an option from households who weigh the risks of drug use correctly, as the evidence suggests they do (Section 2), and gives them back significantly less through the government's budget than it takes.

I argue that the answer turns instead on the children, through three externalities that a parent takes into account only through imperfect altruism towards the child: two that Housing First leaves in place, and one that many believe Treatment First creates. The two that Housing First leaves in place fall on the user's children. Children share their parents' consumption and dwelling but not the drug, so every unit a parent spends on drugs, and every unit of earnings lost while a SUD keeps the parent unemployed, is consumption and housing the child goes without: the *budget externality*. Furthermore, a child whose parent had a SUD while the child lived at home starts adult life with lower productivity: the *earnings externality*. The one that many believe Treatment First creates runs through homelessness. By denying assistance to drug users, Treatment First exposes more of them to the shelter, and a child lives wherever its parent lives: the *childhood homelessness externality*. However, I show that, in fact, Treatment First reduces this externality as well.

Whose children bear these costs depends on who uses drugs. Use is concentrated at the bottom of the income distribution: the average user's income is about 74 percent of the population average, and most households do not use at all. The economics literature has reproduced the two facts together only by letting preferences for the drug differ by education (Greenwood et al., 2025). I model drugs as a luxury good bought at a price, which puts most households at a corner where they consume none. On its own, that would concentrate use at the *top*. What reverses the income gradient is the higher risk of dying due to a SUD. This is because the cost of death rises with income and, unlike losing one's job, is uninsurable. The model therefore explains both facts without assuming systematic differences in preferences.

I proceed with two models of rational forward-looking households. The first is a stylized model with two periods of life, in which a SUD raises the risks of dying and of unemployment. Its sole purpose is to show why the income gradient in drug use reverses. The second is a quantitative general-equilibrium life-cycle model calibrated to the United States. Its purpose is to test whether Treatment First raises or lowers welfare relative to Housing First, and whether moral hazard or the externalities on children drive the answer. Its key ingredients are a housing market and homelessness; rationed, means-tested vouchers; income and unemployment risk, with unemployment insurance and food assistance; retirement and social security; recovery and treatment; serious mental illness (SMI); children who share their parent's consumption and dwelling and start adult life with lower productivity if the parent had a SUD during their childhood; a parent's imperfect altruism towards the child; a drug test that misses some users; a drug price that clears

[3] For the case against Housing First as the federal government now makes it, see the executive order cited above, and on its incentives and costs, Eide (2020); for the case for it, see the National Alliance to End Homelessness, `https://endhomelessness.org/blog/ceo-corner-week-of-july-28/`, and on housing stability and costs, Tsemberis et al. (2004) and Larimer et al. (2009).

the drug market; and a government budget whose net revenue is held at its benchmark level through the probability of receiving a voucher.

I use the calibrated model to study how the economy changes when it moves from Housing First, the benchmark, to Treatment First. Under Housing First, housing assistance is unconditional for eligible households who draw a voucher. Under Treatment First, it is conditional on not using drugs and, for those with a SUD, on being in treatment as well. Testing is imperfect, however, so some users go undetected and keep their housing assistance.

The quantitative model lets me separate the three externalities, moral hazard and general-equilibrium effects from one another. I run the same reform with and without the externalities, with only the screen in effect (Treatment First's drug test, which revokes the voucher of a user it catches) and its savings not spent on vouchers, so the one difference between the two runs is whether drug use harms children. Without externalities, with only moral hazard present, a newborn loses 1.17 percent; with them it gains 2.82 percent, childhood included, and the share of children in a shelter actually falls from 1.03 to 0.95 percent. The parents themselves lose about as much whether there are externalities or not. However, with externalities the children gain more than the parents lose. They consume more at home and are less often homeless, and they start adult life without the earnings penalty, all of which are gains their parents only partly count. In the aggregate, then, there is no trade-off between the externalities: by cutting use, the screen keeps more children out of a shelter than it sends to one by putting their parents out. What the *budget, earnings and childhood homelessness externalities* have to outweigh is only the direct loss to the parent from the removal of an option. As I explain below, even the parent gains once general-equilibrium effects are taken into account.

Breaking down the gains from Treatment First, I show that spending the screen's savings on more vouchers raises the gain to 6.45 percent. Letting the drug price clear as well, as it would after a real reform, brings the gain down to 2.57 percent, since the lower drug price draws some households back into use. In this full long-run general-equilibrium exercise, Treatment First cuts drug use by 21 percent and the share of households with a SUD by 19 percent.

A newborn gains 2.57 percent of lifetime non-housing, non-drug consumption (Section 6.1). Three quarters of it is an externality its parent does not count: 1.50 percent of lifetime consumption that the child gets during childhood, and 0.46 percent from entering adult life without the earnings penalty. The rest, 0.62 percent, has two parts. The first, 0.19 percent, is the part of the child's gains that its parent does count. The second, 0.43 percent, is what the newborn gains later, in its own adult life, like any adult living under Treatment First: a voucher that is easier to draw when sober, and a cheaper drug. The parents lose from the screen's direct impact in partial equilibrium, but once general-equilibrium effects are accounted for, they gain as well, by 0.69 percent of their own lifetime consumption. This is the same adult-life gain as the newborn's 0.43 percent, measured from a different starting point. The parents are already adult, and thus do not discount the gain. A newborn gets it only once their childhood is over, so that adult gain is discounted back to birth and thus is smaller. In general equilibrium, then, both the child *and* the parent gain.

The model captures the micro effects that trials measure on the households they follow, and those with a SUD lose from a sobriety condition in the model too. A trial enrols too few households to change the drug price or the chance of drawing a voucher, so the comparison that matches it keeps both at their Housing First levels. Then – percent of the households that have a SUD at the time of the reform lose, and when both adjust, 67 percent still do. The macro effects the trials do not observe outweigh these micro effects: on the households the condition deters from starting to use drugs, on the vouchers its savings buy for others, and on children across the economy.

Even along the transition, homelessness and childhood homelessness do not rise, and children born at the reform gain 1.45 percent of lifetime consumption. A concern, especially among critics of Treatment First, is that homelessness and childhood homelessness could rise, since the households that have a SUD when the reform lands face losing their vouchers unless they enter treatment and stay sober. These outcomes actually fall rather than rise, for three reasons. First, few households lose their vouchers: one with a SUD keeps its voucher by entering treatment and not using, and the screen catches only 20 percent of users a year. Second, drug use falls from the first year, so more households spend on housing instead of drugs. Third, the screen's savings fund vouchers for other households, and reduce homelessness and childhood homelessness among them.

Children born at the reform still gain, but less than the 2.57 percent of those born in the long run, for two reasons. First, more of their parents still have a SUD, since the share with one falls only as fast as recovery allows, and a child whose parent has a SUD enters adult life with the earnings penalty, whether or not the parent is in treatment. Second, they grow up before the chance of drawing a voucher has fully risen, since part of what pays for the extra vouchers is the taxes of adults who grew up without that penalty, and those adults are not yet working.

2 Related literature

Randomised trials of Housing First. A large empirical literature evaluates programmes that provide rapid housing without conditioning on sobriety, and finds large and persistent improvements in housing stability and reductions in shelter and other public-service use (Tsemberis et al., 2004; Padgett et al., 2006; Larimer et al., 2009; Goering et al., 2014; Aubry et al., 2016). Evidence on substance use is mixed: most studies find no statistically significant change, and some report modest reductions in alcohol-related outcomes. Costantini (2025) finds that enrolling in Housing First lowers the mortality of homeless veterans with mental illness, while the Treatment First programmes of the Department of Veterans Affairs have no lasting effect on it; Section 5.2 compares the model with this finding. The one trial of homeless families, HUD's Family Options Study, followed 2,282 families for three years: an unconditional subsidy reduced later homelessness, family separations and the children's school absences and behaviour problems (Gubits et al., 2016, 2018).

These studies, however, measure outcomes over six to thirty-six months, which is short against a recovery process that Vaillant (1995) documents as gradual and protracted. A further limitation of a trial, and perhaps a deeper one, is its scope. A trial randomises households within a fixed environment, and so leaves out exactly the things that decide the welfare comparison: the impact on the behaviour of households outside the sample, the impact on the government's budget, and, most importantly, the effects on the participants' children. No trial of a homelessness programme follows all households, or follows the participants' children into adult life, and no trial randomises what a parent's drug use does to a child at home; that evidence is observational (see the paragraph on intergenerational transmissions below).

Deterrence effects of zero-tolerance policies. The closest evidence on how much a drug test with a severe penalty deters drug use comes from the US military, which in 1981 began testing its personnel at random, 10 to 20 percent of a unit every month, and dismissing those who failed. Mehay and Pacula (1999) use this policy change to compare service members with civilians, following the same people over time. They find that, among service members who stayed, past-year marijuana use fell by about 61 percent compared with what it would have been without testing. Section 6 uses this response to set how often Treatment First's screen catches a user.

Economic models of drug use. Modelling substance use as forward-looking optimisation dates to Becker and Murphy (1988), and the assumption is empirically defensible at the margin where policy operates. Use responds to its price, even for drugs as habit-forming as heroin and cocaine, and more in the long run than in the short run, the signature of forward-looking choice rather than myopia (van Ours, 1995; Grossman and Chaloupka, 1998; Saffer and Chaloupka, 1999; Caulkins, 2001; Gallet, 2014). Greenwood et al. (2025), who allow perceived and true SUD risk to differ, find that the model-implied beliefs are actually very close to the true risk. Moreover, young people, if anything, overestimate the risk of becoming dependent, and those who perceive a higher risk drink less (Lundborg and Lindgren, 2002). A separate literature emphasises temptation and dynamic inconsistency (Laibson, 1997; Gruber and Kőszegi, 2001), under which present-biased households over-consume relative to their own long-run preferences. With such assumptions, Treatment First has a rationale even with no externalities at all, since a present-biased household could value a sobriety requirement as a commitment device. Thus, adding dynamic inconsistency to the model would strengthen the case for conditionality. The paper also departs from Greenwood et al. (2025) in how it generates the income gradient in drug use. There, use concentrates among the less educated because their preferences for drugs differ; here no preference parameter varies with income, and the gradient comes from

the uninsurable mortality risk of a SUD (Section 3).

General-equilibrium models of homelessness. I build directly on İmrohoroğlu and Zhao (2022), who model homelessness as a rational response to income risk, mental-health shocks, housing costs and housing policy (see also Abramson, 2026, on how eviction policy changes who becomes homeless, and Rotberg and Zhang, 2024, on rent control's effect on homelessness). In their framework, the effect of housing policy on homelessness is not tied to drug use, and thus a policy that requires sobriety has no margin to affect. My model includes that margin by adding endogenous drug use, disorder and recovery, along with mortality that depends on a household's state.[4]

Intergenerational transmissions. The *earnings externality* depends on the connection between a parent's SUD and its child's adult earnings. Parental drug abuse is recorded in more than a third of removals into foster care (Adoption and Foster Care Analysis and Reporting System (AFCARS), FY2017), household substance abuse is one of the adverse childhood experiences of Felitti et al. (1998), and children of a parent with a SUD are more likely to acquire one themselves (Mellentin et al., 2016; Yoon et al., 2013). A parent who stays unemployed longer and spends on the drug has less to invest in the child, and that investment shows up in adult earnings: family income raises children's achievement (Dahl and Lochner, 2012), parental investment in education accounts for roughly half of the intergenerational correlation in earnings (Restuccia and Urrutia, 2004), and a year of schooling returns around ten percent (Card, 1999).

Nilsson (2017) is, to my knowledge, the only study that measures, with a natural experiment, how a parental substance shock affects the child's adult earnings. Since this is the best available evidence, the model's productivity penalty is set to its estimate (Section 5), even though the study's shock, exposure to alcohol in the womb, does not map one to one into my model, in which a parent's disorder after the child is born affects the child's productivity. The size of the penalty can also be checked indirectly, by comparing model outcomes to estimates in the literature. In the model, the odds that a child whose parent has a SUD develops one themselves in adulthood are at the low end of the empirical evidence (Section 5.2). This suggests that, if anything, the penalty I use is at the low end as well.[5]

The actual experiment the Nilsson paper uses is as follows. Two Swedish counties allowed grocery stores to sell strong beer for a year in 1967 and 1968. What Nilsson found was that children conceived in the first months of the experiment by mothers under 21 earned 24.4 percent less at age 32 than the cohorts born just before and after them, as well as the same cohorts elsewhere in Sweden. Moreover, the loss persisted beyond age 32.

There are two key points to highlight about this study. First, the penalty estimated in Nilsson is an average over the whole exposure group. Specifically, every child a young mother in the two counties conceived during those months counted as exposed, whereas only some of those mothers drank more. Second, the earnings loss is persistent, and this is how I model it. Non-experimental evidence agrees. For example, adult children of problem-drinking parents are more likely to be unemployed and earn less (Balsa, 2008). I do not aim in this paper to offer another estimate, but rather take those as given and price the effects inside the model.

Homelessness persists across generations too (Bassuk et al., 1997). In the model, this persistence comes from the persistence of earnings, not from homelessness itself. The intergenerational elasticity of earnings in the United States runs from about four tenths (Solon, 1992) to about six tenths (Mazumder, 2005), with the same persistence in ranks across the distribution (Chetty et al., 2014). A poor environment does lower a child's adult earnings beyond that (Chetty et al., 2016; Chyn, 2018), but the model imposes no penalty for growing up in a shelter. However, since Treatment First lowers both homelessness and childhood homelessness, and does not raise them even in the short run (Section 6.4), adding such a penalty would only strengthen the case for Treatment First.

[4] The paper does not focus on the disamenity that homelessness may impose on the rest of the population. In a contingent valuation, Loubière et al. (2020) find that Europeans would pay about 0.2 percent of average earnings a year to end homelessness. I also solved the model with a disamenity of that kind, and it barely changes the results. The welfare numbers in the paper leave it out; with it, at that value, no long-run gain in Table 5 moves by more than 0.14 percentage points.

[5] Even when the penalty is set to zero, Treatment First results in welfare gains.

Parental altruism towards children. How much of these costs a parent counts depends on the weight it puts on its child. Altonji et al. (1997) reject perfect altruism: when a dollar moves from a grown child to its parent, the parent gives back at most thirteen cents of it. In the model a parent and its grown child do not transfer to each other, so that test cannot be run in it. Instead, the weight is set by behaviour. In the 2024 National Survey on Drug Use and Health (NSDUH), unmarried adults living with one other adult and no child are 17.5 percent more likely to use drugs than lone parents of the same age, sex and family income who have one child. The weight is the one at which removing a parent's concern for its child raises drug use in the model by as much (Section 5). The externality is then the share of what its choices cost the child that the parent leaves out: one minus that weight.

That a drug user has children it affects is supported by the data. Drug users are nearly as likely as others to be parents: in the National Survey of Family Growth 2017–2019, 75 percent of adults aged 40 to 49 who used hard drugs in the past year had a child, against 81 percent of those who used none. They are only somewhat less likely to live with it.[6] Among adults aged 26 to 49, 40 percent of those with a drug use disorder have a child under 18 at home, against 48 percent of comparable adults who used no illicit drug. That gap is too small to matter: even if all of the gain from Treatment First that comes in childhood shrank in proportion to it, the gain would be only 11 percent smaller.[7]

The hazards and the drug market. Each of the model's hazards is anchored on, or checked against, a clinical or epidemiological estimate: the excess mortality of a SUD, of homelessness and of SMI (Plana-Ripoll et al., 2019; Heiberg et al., 2018; Meyer et al., 2023; Olfson et al., 2015), the onset of SMI and relapse under dual diagnosis (Nielsen et al., 2017; Bradizza et al., 2006), recovery (Vaillant, 1995; Milby et al., 2005), and the employment cost of losing housing (Desmond and Gershenson, 2016). Section 5 takes them one at a time. Making the drug price endogenous lets a policy that changes the number of users change what using costs everybody else. The model's demand elasticities are not calibrated and are compared with the estimates of Saffer and Chaloupka (1999), Grossman and Chaloupka (1998) and Olmstead et al. (2015) in Section 5.2; its supply elasticity is set from Mace et al. (2020) in Section 5.

3 A stylized model

In the data, fewer than one in ten adults have a SUD, mild included, which is what this paper counts as use, and their use is concentrated among people with low incomes: the average user's income is 0.74 of the population average in the 2024 NSDUH. This section shows how both facts can be explained inside a simple economic model of drug use. There are two key ideas. Drugs are a luxury good, so most households consume none of them. And a SUD carries a risk of dying that cannot be insured against, a loss that grows with income, which deters high-income households and not low-income ones and so reverses the income gradient in drug use.

3.1 The model

There are two periods, adulthood and old age, and a continuum of households. A period stands for thirty years of adult life, so the two of them span ages twenty-six to eighty-five. Parameters with a natural annual counterpart are compounded to that length: the discount factor is β per period and $\bar{\phi}$ is survival over thirty years.

A household is characterised by its productivity y, fixed over its life and lognormally distributed across households, $\log y \sim \mathcal{N}(-\frac{1}{2}\sigma_y^2, \sigma_y^2)$, normalised so that its mean is one. Everyone works in the first period. In the second, a household is out of work with probability u_0, or $u_1 > u_0$ if it has a SUD. With $e = 1$ for a

[6] Users who have a child are less likely to live with it, partly because the state removes children: parental drug abuse is recorded in more than a third of removals into foster care (AFCARS, FY2017), and foster care is itself a very poor environment for a child.

[7] The model likely overstates childhood homelessness, since every homeless parent in the model takes its child into the shelter. The overstatement comes from fathers: a homeless mother's child usually goes with her, but shelters for adult males do not take children, so a homeless father's child usually stays with its mother or other relatives, and some children are taken into foster care.

household that works and $e = 0$ for one that does not, a household's earnings are $y\,e$: households out of work earn nothing, and there is no unemployment insurance. What a household can do is save, at zero interest, in the first period; it cannot borrow.

The one transfer is food assistance, paid on each period's earnings as

$$F(y\,e) \;=\; \max\{0,\; \bar{c} - \varrho\, y\, e\}, \tag{1}$$

where $\bar{c}$ is the maximum benefit and ϱ the share of earnings by which it is reduced. Food assistance is in the model for one reason. To show the income reversal, Section 3.3 needs to solve the model with saving switched off. A household out of work would then have nothing to consume, which with $\sigma > 1$ leaves its problem undefined; food assistance bounds its consumption from below.

Consumption c and drug consumption v form the bundle

$$X(c, v) = c^{1-\theta}(b + v)^{\theta}, \tag{2}$$

and flow utility is

$$u(c, v) \;=\; U\big(X(c, v)\big) \;+\; \aleph, \qquad U(X) = \frac{X^{1-\sigma}}{1-\sigma}. \tag{3}$$

The weight on the drug θ is its share of the whole, and σ is the coefficient of relative risk aversion.[8] Drugs can be bought only in the first period, at the price p_v. θ and b govern the decision of whether and how much drugs to consume. The marginal utility of drugs is proportional to $\theta/(b + v)$, which means that θ mostly determines how much a user uses, whereas b affects the decision to use at all. When $b > 0$, the marginal utility of the first unit is finite, so some households do not use. When $b = 0$, the marginal utility from the first unit is infinite and thus every household uses, regardless of the consequences. Finally, $\aleph$ denotes the flow value of being alive unrelated to consumption, and this is what gives life a positive value even when consumption is low.

Using drugs carries the risk of a SUD ($d = 1$ if a household has one and $d = 0$ if not), the same whatever the quantity,

$$\pi(v) \;=\; \pi_d\, \mathbf{1}\{v > 0\}. \tag{4}$$

The whole risk therefore sits on the decision to use at all, not on how much a user uses.[9] Every household enters the first period without a SUD. The SUD is realised at the end of it, and the household survives into the second with probability

$$\phi(0) \;=\; \bar{\phi}, \qquad \phi(1) \;=\; \bar{\phi}^{\,\kappa_{\phi d}}, \tag{5}$$

where a SUD lowers the probability of surviving the period by raising it to the power $\kappa_{\phi d} > 1$. The second period is the last and has no drug-use decision, so to a household with savings a and SUD status d it is worth

$$W(a, y, d) \;=\; (1 - u_d)\, U\big(X(y + a + F(y), 0)\big) \;+\; u_d\, U\big(X(a + \bar{c}, 0)\big) \;+\; \aleph. \tag{6}$$

In the first period, a household that does not use chooses its saving to maximise its lifetime utility,

$$V^0(y) \;=\; \max_{\{a \geq 0\}} \Big\{ U\big(X(y - a + F(y), 0)\big) + \aleph + \beta\, \bar{\phi}\, W(a, y, 0) \Big\}, \tag{7}$$

and one that uses chooses its saving and its drug consumption to maximise it,

$$V^1(y) \;=\; \max_{\{a \geq 0,\; v > 0\}} \Big\{ U\big(X(y - a + F(y) - p_v v,\; v)\big) + \aleph + \beta \Big[(1 - \pi_d)\, \bar{\phi}\, W(a, y, 0) + \pi_d\, \bar{\phi}^{\,\kappa_{\phi d}}\, W(a, y, 1) \Big] \Big\}. \tag{8}$$

It uses if $V^1(y) > V^0(y)$. A SUD reaches the second period through two channels: the mortality channel, which lowers the chance of living it, and the unemployment channel, which raises the chance of living it unemployed.[10]

[8] All the qualitative results of this section hold with log utility as well, with the model recalibrated to the same targets.

[9] The results hold even when I let that probability rise with use.

[10] The results in Section 3.3 also hold in a version of this model with housing, a shelter, unemployment insurance and housing vouchers. There, if a SUD has no consequences, the average user earns 3.54 times the population average; the risk of dying, on its own, takes this to 0.52 at twice its calibrated strength, while the risk of losing one's job, on its own, leaves it at 3.54. In contrast, after switching off the savings option, the risk of losing one's job brings it down to 0.56 on its own.

3.2 Calibration

Here, I calibrate the model. Table 1 lists the parameters I assign externally and Table 2 lists the ones I assign internally. All five internally fitted parameters are determined jointly, but each is calibrated based on the moment it affects the most. The model hits all the moments exactly. I created a replication package (`data/DATA_APPENDIX.md`) which provides every target, and the dataset used to compute it (with its year and universe). Below I give each moment and its source.

3.2.1 External calibration

Preferences and demographics are conventional: the discount factor over a thirty-year period is $\beta = 0.86$, which is 0.995 a year compounded,[11] and risk aversion is $\sigma = 2$, as in the quantitative model. Survival over a period, the cumulative chance of surviving from age twenty-six to age fifty-five, is $\bar{\phi} = 0.93$, from the life table for the total population of the United States in 2010 (Arias, 2014). The chance that use causes a SUD, $\pi_d = 0.41$, is set so that the ratio of the SUD rate to participation matches the data in the 2024 NSDUH, 3.68 against 8.90 percent of adults aged 26 and over.

The dispersion of log productivity σ_y is set so that the poorer half of the lognormal holds 17.2 percent of total income, the share of earnings held by the poorer half of households aged 26 to 65 with positive wage and salary income in the 2022 Survey of Consumer Finances (SCF). In the second period, a household without a SUD is out of work with probability u_0, 4.9 percent, the average unemployment rate from 2014 to 2024, and one with a SUD with probability u_1, 16.2 percent, the unemployment rate of adults aged 26 and over with a SUD in the 2024 NSDUH. Food assistance tapers at $\varrho = 0.30$, the benefit reduction rate of the Supplemental Nutrition Assistance Program (SNAP), and the price of a unit of drugs p_v is normalised to one.

3.2.2 Internal calibration

The flow value of being alive $\aleph$ is matched to a value of a statistical life (VSL) of 80 times average annual income (Viscusi and Aldy, 2003).[12] The parameter b, which sets the marginal utility of the first unit of drugs, is chosen to match a drug participation rate of 8.90 percent, the share of respondents aged 26 and over in the 2024 NSDUH with a drug use disorder of any severity. The weight on drugs θ is set to match the aggregate drug spending over the aggregate income of users, 16.0 percent, from retail spending in Midgette et al. (2019) and the incomes of users in the 2024 NSDUH. The level of food assistance $\bar{c}$ is set so that total food assistance is 0.42 percent of total labour income, the ratio of SNAP outlays to gross domestic product (GDP).

The mortality exponent $\kappa_{\phi d}$ is set to match the average income of users, 0.74 of the population average in the 2024 NSDUH, because the death risk is what decides how far down the income distribution use sits, as Section 3.3 shows.

3.3 Who uses drugs

Drugs in the model are a luxury good bought at a price, so a household with twice the income gives up less utility per dollar spent on them. Without consequences to using drugs, drug use would concentrate at the top of the income distribution. With b recalibrated so that participation stays at 8.90 percent, the average user would earn 3.51 times the population average. I spend the rest of this section showing that it is the uninsurable risk of dying that puts drug use at the bottom of the income distribution.[13]

In Figure 1, I vary the strength of each of the two penalties, mortality and unemployment. I start from zero and run up to 200 percent of each of their calibrated values, keeping the other penalty switched off. It

[11] The annual discount factor here is higher than the one in the quantitative model, and this is deliberate. A period in this model is thirty years, which means that every consequence of using drugs arrives only thirty years after the decision. In the quantitative model, in contrast, the consequences of drug use arrive more frequently. With a low annual discount factor in the stylized model, the risk of death many years into the future therefore counts for much less, and the mortality channel would be dominated by impatience, which the model is not about. I therefore use a higher factor to correct for coarseness of the horizon.

[12] No household has a negative value of life in period 1, so death is never an optimal choice.

[13] A household can buy life insurance, but it protects its descendants, not the household itself or the value of its own life.

Table 1: Externally set parameters

Parameter	Description	Source or rationale	Value
β	Discount factor over a thirty-year period	0.995 annually, compounded	0.86
σ	Risk aversion	Standard; Attanasio and Weber (2010)	2
$\bar{\phi}$	Survival over a period, no SUD	US life table 2010, Arias (2014), ages 26–55	0.93
π_d	Chance that use starts a SUD	NSDUH 2024: SUD rate over participation	0.41
σ_y	Dispersion of log productivity	SCF 2022: bottom half holds 17.2% of earnings	0.95
u_0	Unemployment rate in period 2, no SUD	BLS CPS, average 2014–2024	0.049
u_1	Unemployment rate in period 2 with a SUD	NSDUH 2024, ages 26 and over	0.16
ϱ	Food assistance taper	SNAP benefit reduction rate	0.30
p_v	Price of a unit of drugs	Normalisation	1.00

Table 2: Internally calibrated parameters

Parameter	Target moment	Data	Model	Value
$\aleph$	VSL over income	80	80.0	8.64
b	Drug participation, NSDUH 2024	8.9%	8.9%	0.032
θ	Drug spending over users' income	16.0%	16.0%	0.22
$\kappa_{\phi d}$	Users' income over the average, NSDUH 2024	0.74	0.74	3.56
$\bar{c}$	Food assistance over labour income	0.42%	0.42%	0.10

scales $\kappa_{\phi d} - 1$ for the mortality channel and $u_1 - u_0$ for the unemployment channel. I recalibrate b each time to keep participation at its calibrated value. This means that the share of households that use is the same across the figure, and what changes is only who uses.

The left panel shows the calibrated model in which households can save. The mortality channel alone takes the average user from 3.51 times the population average down to 0.64 at its calibrated strength, crossing the average at 73 percent of that strength, and to 0.27 at 200 percent.[14] In contrast, the unemployment channel alone leaves the average user at 3.51 times the population average even at 200 percent: on its own it does not change who uses at all.

The right panel shows that when the precautionary savings are switched off both the risk of dying and the risk of losing one's job reverse the income gradient. Without savings, the unemployment channel on its own takes the average user to 0.75 times the population average at its calibrated strength, and to 0.27 at 200 percent.

Who uses depends on two things that vary with income. The first is the cost of acquiring drugs, and the second is the penalty drugs inflict. The utility cost of the drug falls with income, because a household with more income gives up less marginal utility for each dollar it spends on drugs. Therefore, ceteris paribus, drugs cost a rich household less than a poor one.

The penalty inflicted by drugs does not rise with income when it can be partially insured against, but does rise when it cannot be. A richer household can save more to protect itself in the event of unemployment. Thus, its loss is roughly proportional to its income. As a result, in utility terms, the cost of losing its employment is about the same as the cost imposed on a poor household. Death, on the other hand, cannot be insured against. Dying eliminates the full continuation value of being alive, and that loss grows with income, so it deters high-income households but not low-income ones. Without the ability to save, losing a job takes a household down to $\bar{c}$ whatever it earned, so that loss too grows with income, which is why in the right panel the unemployment channel reverses the gradient as well.

4 The quantitative model

This section describes the model the paper's numbers come from. It keeps the preferences of the stylized model of Section 3 and the two risks a SUD brings there, and adds what that model leaves out: housing, homelessness and vouchers; annual periods; unemployment insurance; income risk during working age;

[14] With both channels at their calibrated strength, the average user earns 0.74 times the population average, as in the data.

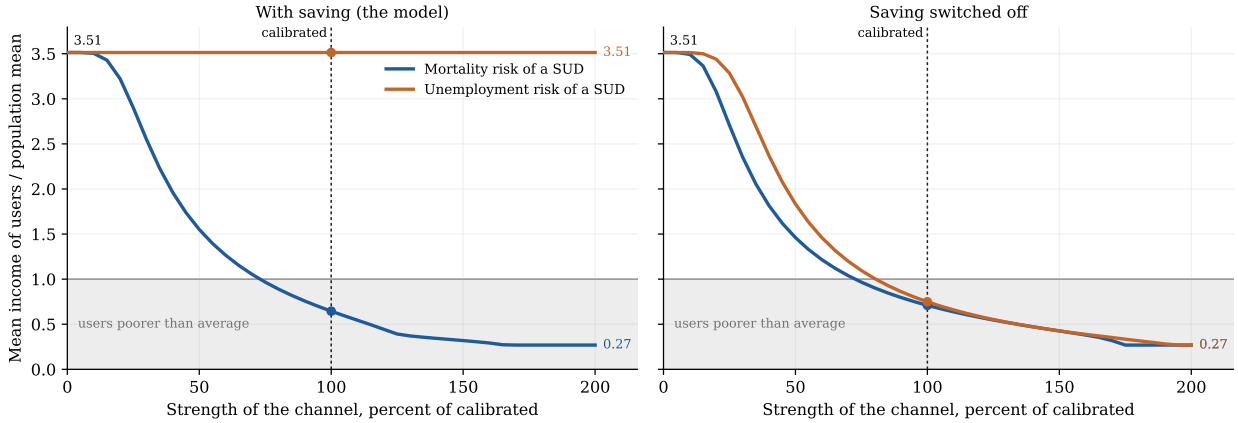


Figure 1: What it takes to reverse the income gradient in drug use

Notes: each curve scales one channel with the other switched off. Left: the model. Right: the same parameters with saving switched off. The vertical axis is average income of users over the population average, and b is recalibrated at every point so that participation is the calibrated 8.90 percent. The vertical dotted line marks the calibrated strength.

retirement; progressive income taxes; interest on savings; a drug-use decision in every year; recovery and treatment; SMI; a higher risk of dying, worse employment prospects and a lower chance of recovering while homeless; children who live with their parents; a drug test that misses some users; a government budget held at its benchmark net revenue; and an endogenous drug price. Notation follows Section 3 wherever the two models share an object: θ is the drug weight, σ the risk aversion, b the Stone-Geary shifter, $\aleph$ the flow value of being alive, π_d the chance that use starts a SUD, $\kappa_{\phi d}$ the mortality multiplier for the disorder, p_v the drug price and β the discount factor. The values are in Section 5.

4.1 Demographics and the state and choice sets

At model age $j = 1$, a child becomes an adult, leaves its parent's home, joins the workforce and forms its own household, with a child of its own. The household lives at most J annual periods, works while $j < J_R$ and is retired from J_R. Its child lives with it for the first J_K years and then leaves to form its own household in turn.

Each period a cohort of unit mass enters at $j = 1$, made up of the children who leave home that period, at their parent's death if that comes before J_K and at J_K otherwise (equation (29)). It replaces the mass of the households who died, so the mass of households is constant in a stationary equilibrium. The individual state is age j together with

$$s = (a,\, y,\, e,\, d,\, m,\, \eta) \in \mathbb{R}_+ \times \mathcal{Y}(j) \times \{1,2,3\} \times \{0,1,2\} \times \{0,1\} \times \{0,1,2\}, \tag{9}$$

where a denotes assets, y productivity, e employment status, d the SUD, m SMI and η the household's standing with the voucher programme. The SUD takes three values because the parent's valuation of its child's prospects depends on the history of the disorder and not only on its present (Section 4.10): $d = 0$ denotes a household that has never had the disorder, $d = 1$ refers to one that had it and has recovered, and $d = 2$ stands for one that has it now.[15] The voucher state is $\eta = 0$ for a household without a voucher, $\eta = 1$ for one that holds one, and $\eta = 2$ for one excluded from the programme after failing Treatment First's condition (Section 4.9).

Each period the household chooses the bundle

$$\left(h,\, v,\, \tau,\, a'\right) \in \{h_1, h_2, h_3, h_4\} \times \mathbb{R}_+ \times \{0,1\} \times \mathbb{R}_+, \tag{10}$$

[15] Section 3 wrote $d \in \{0,1\}$ for the disorder absent and present; here $d = 1$ is a household that has recovered, since the quantitative model keeps the history.